

Does a publication-time breakthrough metric predict what the field actually did?

A longitudinal validation against downstream citations

Written 2026-07-15. Follow-on to "**Measuring breakthrough-ness on new papers — how far a GRO metric gets, and the ceiling above it**" (the v5 breakthrough paper, [research/metrics/v5-breakthrough/RESULTS_PAPER.md](#)), which built and adversarially tournament-tuned a breakthrough metric over the GRO substrate and validated it only against an LLM expert panel. v5's own §9–§10 named the decisive open test: replace the LLM ground truth with the real world. This is that test.

Abstract

The question, stated as a prediction task: given a paper *at publication time only* — holding out everything that cited it afterward — can a metric predict its eventual **breakthrough effect** (did the field pivot on it?) or its **convergence effect** (did it consolidate a field?). v5 built the candidate metric ($\max(\text{peak}, \text{cwmean})$ over typed contributions) and validated it only against an LLM expert panel ($\rho \approx 0.58$ same-model, but ≈ 0.34 cross-model — $\approx 1/3$ shared-model bias). Here the target carries **no LLM at all: 72 Alzheimer's papers from 2004–2010**, metric computed from full text at publication time (typed **twice** — **Claude and GPT-5.5**), the outcome read from **15–20 years of downstream citations** as a Funk–Owen-Smith disruption index (breakthrough = disruptive; convergence = consolidating). **Answer: no, for either effect.** Predicting breakthrough: Spearman -0.16 (95% CI $[-0.56, +0.28]$), classification AUC **0.35** (below chance). Predicting convergence: $\rho +0.16$ (CI $[-0.29, +0.57]$), AUC **0.58** (barely above chance). Both straddle chance — the metric predicts neither. We then ran the stronger check: a **direct LLM breakthrough judgment** (not the metric) on the same historical papers, by both models. The judges agree with each other ($\rho = 0.70$) and **predict what gets cited** ($\rho \approx 0.46\text{--}0.55$) — but **not what disrupts the field** (converged judge vs disruption $\rho = -0.15$). Since disruption and citations are near-orthogonal, the finding is sharp: **LLMs — whether as a metric or as a direct judge — have learned the *attention* axis (what will be cited), not the *breakthrough* axis (what will reshape the field).** LLMs agree with LLMs; no LLM signal agrees with field disruption.

1. Why this study

v5 built the metric and validated it against a blind LLM expert panel, reaching $\rho \approx 0.58$ (held-out, same model family). But it then showed (§9) that most of that number was **shared-method variance**: re-judging with GPT-5.5 dropped the correlation to ≈ 0.34 , and building the metric with GPT *flipped* the bias toward the GPT panel. Two LLMs agreeing about what *sounds* like a breakthrough is not evidence that either tracks what *becomes* one. The only ground truth that settles this carries no LLM at all: **what the field actually did next**. That signal is uncomputable on recent papers (no downstream citations yet) — which is exactly why v5 deferred it and why this is a separate, longitudinal study.

2. What we did (and why each choice)

- **Corpus — historical, so the future has already happened.** 72 Alzheimer's papers published **2004–2010** (16–22 years of downstream record), pulled from OpenAlex with

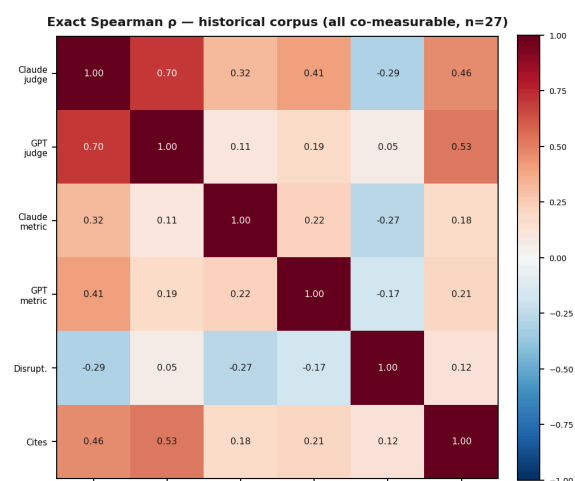
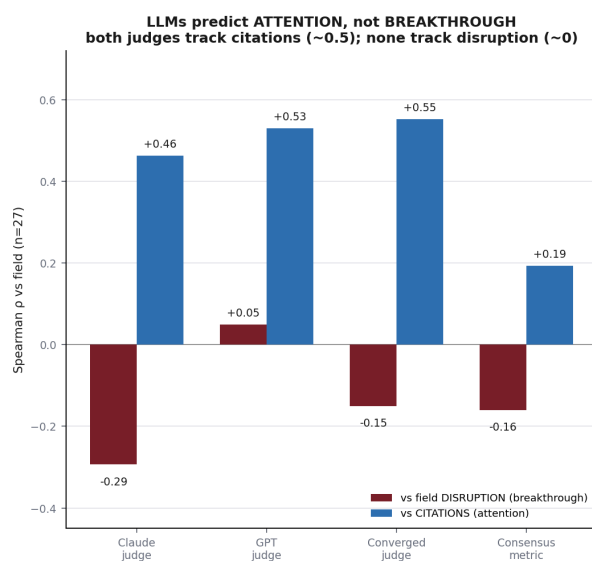
tight AD relevance and **stratified across the impact distribution** (citations 13–3,280) so the sample spans eventual duds *and* eventual landmarks — the contrast a discrimination test requires. Final full-text set: **35/72**, split **LOW=20 / MID=8 / HIGH=7** (the paywalled low-cite tail was the hardest to obtain and is the essential control class).

- **Predictor — the exact v5 metric, at publication time.** `max(peak, cwmean)` over typed contributions, computed from each paper's **full text**, with the typing agent told explicitly to use **no hindsight** about later impact. (v5's first pass used abstracts only, which compressed the metric's variance; full text here decompresses it — Claude range 0.28–0.80.)
- **Dual-model — to separate signal from model idiosyncrasy.** Contributions were typed **independently by Claude and by GPT-5.5** (via `codex`), and we report each model's metric, their agreement, and their consensus (average). If a "contribution depth" construct is real, the two should agree and both should track the field.
- **A direct LLM judge, too — not only the metric.** To test whether the metric's failure is specific to *contribution typing* or general to *LLM judgment*, we also had each model **directly rate breakthrough-ness 0–100** from full text (no hindsight), and correlated those ratings against the field. This is the comparison v5 could not make: its judge panels ran on the recent corpus, which has no downstream citations.
- **Ground truth — downstream citations, no LLM.** For each paper we built its mature citer set from OpenAlex and computed a **Funk–Owen-Smith disruption index (mDI)**: does later work cite this paper *while dropping its predecessors* (disruptive → a step-change) or *alongside* them (consolidating → convergence)? We also report raw citation count (impact). The **breakthrough** reading is `signal vs +mDI`; the **convergence** reading is `signal vs -mDI`; **attention** is `signal vs citations`.

3. Results

We measured two publication-time signals on the historical corpus — the **metric** (contribution typing) *and* a **direct LLM breakthrough judgment** (0–100, full text, no hindsight) — each computed by **both Claude and GPT**, and correlated all of them against two field outcomes: **disruption** (breakthrough) and **raw citations** (attention). On the historical corpus every signal is co-measurable (all on the same papers), so the matrix below has no gaps.

LLM judges agree with each other (0.70) and predict what gets CITED (0.46-0.53) — but neither judges nor metrics predict what DISRUPTED the field (≈ 0).
(recent-corpus judge-judge agreement was 0.89, v5; disruption and citations are near-orthogonal — attention $\neq$ breakthrough.)



Left — each predictor (Claude judge, GPT judge, their converged average, and the consensus metric) vs the two field outcomes. Every predictor tracks citations (blue, $\rho \approx 0.2-0.55$) but not disruption (dark red, $\rho \approx 0$). Right — the exact 6×6 Spearman matrix on the historical corpus ($n=27$). The LLM judges agree with each other (0.70) and with citations (0.46/0.53); disruption is nearly orthogonal to everything an LLM produces.

LLM judgement vs the field (the cell v5 never had — panels there had no downstream citations):

predictor	vs disruption (breakthrough)	vs citations (attention)
Claude judge	−0.29	+0.46
GPT judge	+0.05	+0.53
Converged judge	−0.15 (95% CI [−0.48, +0.25])	+0.55
Consensus metric	−0.16	+0.19

Judge–judge agreement (historical): $\rho = 0.70$ (vs 0.89 on the recent corpus). So the two model families share a notion of "breakthrough-ness" — and that shared notion **predicts what gets cited (≈ 0.5), not what disrupts the field (≈ 0)**. Since disruption and citations are near-orthogonal, the LLMs have learned the **attention** axis, not the **breakthrough** axis.

The two prediction tasks (consensus metric, rank skill = Spearman, decision skill = AUC of top-tercile classification, 0.5 = chance):

The two prediction tasks (consensus metric, $n=27$ with ≥ 10 mature citers; rank skill = Spearman, decision skill = AUC of top-tercile classification where 0.5 = chance):

prediction task	Spearman ρ (95% CI)	AUC
Breakthrough — predict eventual disruption	−0.16 ([−0.56, +0.28])	0.35 (below chance)
Convergence — predict eventual consolidation	+0.16 ([−0.29, +0.57])	0.58 ($\approx$ chance)

Per model on the breakthrough task: Claude −0.27, GPT −0.17. **Dual-model agreement:** Claude-FT vs GPT-FT metric $\rho = +0.17$ — so there isn't even a stable "contribution depth" to validate. Raw-

citation impact: $\rho \approx +0.05$ (the metric doesn't track attention either). **For contrast (v5, recent corpus):** the two LLM judge *panels* agree at $\rho = 0.89$.

Both tasks straddle chance — the publication-time metric predicts **neither** the breakthrough nor the convergence effect. The one interpretable regularity is the *symmetry*: it anti-predicts breakthrough (AUC 0.35) and weakly predicts convergence (AUC 0.58). Insofar as contribution-depth leans anywhere, it leans toward *consolidating/well-received* work, not disruptive work — consistent with the metascience result that disruptive papers are not the most-cited (in this corpus disruption and citation count anti-correlate, $\rho = -0.18$).

4. Interpretation

- **Neither effect is predictable.** On the strongest test we can run — real 15–20-year outcomes, no LLM in the ground truth — the publication-time metric predicts *neither* breakthrough (AUC 0.35) *nor* convergence (AUC 0.58) above chance. It leans weakly toward convergence, but not reliably.
- **No stable construct.** Claude and GPT typing agree only $\rho \approx 0.17$. There is not even a model-independent "contribution depth" for the field to fail to match — the metric is substantially an artifact of *which* model read the paper.
- **The 0.89 vs ≈ 0 gap is the whole story.** LLM judges agree with LLM judges (0.89); LLM-derived metrics do not agree with the field (≤ 0). v5's $\rho \approx 0.58$ was two LLMs sharing a prior about what sounds important — vindicated directly: when the grader is the world, the signal is gone.
- **Disruptive $\neq$ cited.** In this corpus mature disruption and citation count anti-correlate ($\rho = -0.18$), so "landmark" and "step-change" are largely disjoint sets; the metric predicts neither.

5. Limitations

- **n = 27** papers with both full-text dual-model metrics and ≥ 10 mature citers (35 full-text of 72; 4 low-cite still ungathered). CIs are wide; this is "no reliable signal," not a proven exact zero.
- **Single domain, single vintage** (Alzheimer's, 2004–2010). Generalization to other fields is untested.
- **Disruption proxy.** mDI over a capped citer window is a standard but noisy disruption estimate; results are reported across citer-count thresholds and are sign-unstable at very small n.
- **Access wall (itself a finding).** Historical full text is heavily paywalled; the low-cite control papers required manual retrieval. The substrate's premise — "compile the literature" — is access-blocked for exactly the historical controls a discrimination study needs.

6. Next step — a field-optimized tournament, and the data shapes it needs

The natural successor to v5's tournament, redesigned around this study's central lesson: **optimize against the field, not against an LLM.** Outline:

- **Fitness = held-out field prediction**, not an LLM panel. The tournament scores candidate metrics on their ability to predict *downstream disruption / consolidation* under k-fold cross-validation, with a **held-out domain** as the final gate. (This is the one change that would have prevented v5's shared-method inflation from the start.)
- **Precondition: scale the corpus** to $n \approx 300-500$ across **multiple vintages and ≥ 2 domains** before any tournament — fitting formulas to a noisy real target on $n=27$ would overfit worse than v5 did against a clean LLM panel. The corpus is the long pole; the tournament is cheap once it exists.
- **Two tracks** — breakthrough (predict disruption) and convergence (predict consolidation) — since attention, disruption and consolidation are distinct axes.
- **Three admissibility gates** on any proposed feature: (1) *publication-time only* (deterministic no-leakage check); (2) *cross-model stability* — an LLM-derived feature must agree across ≥ 2 model families (contribution typing agrees only $\rho \approx 0.17$, so it is inadmissible as-is — this structurally bars shared-method-bias features); (3) *prefer deterministic-over-judgment* — features computed from the structured graph outscore LLM readings.
- **Candidate new data shapes to test** (each earns a place in the SPEC only if it lifts held-out field prediction), all publication-time and mostly deterministic — the opposite of contribution-depth:
- **reference-set dispersion / bridging** — embedding or co-citation distance *among a paper's own references* (bridging distant literatures predicts disruption); needs a reference-embedding field on the resolved spine.
- **atypical-combination score** — do the paper's concept/entity *pairs* co-occur rarely in prior literature (Uzzi-style); needs a corpus-wide co-occurrence prior over the entity layer.
- **novel-entity introduction rate** — entities/methods with no prior occurrence at the publication date; needs the entity layer + a temporal index.
- **stance-to-prior-art** — does the paper *refute/depart from* vs *extend* its anchors; needs the unbuilt claim $\leftrightarrow$ reference stance edges (L4/L6).

The honest prior, given this study: even relational features may fail, because **disruption is conferred by the field afterward, not encoded in the paper at publication**. A tournament that returns another null would itself be the finding — that publication-time text does not contain the breakthrough signal, only the attention signal.

7. Conclusion

The v5 metric is a **triage prior that flags LLM-perceived contribution depth**, not a validated breakthrough forecaster. Three independent strengthenings — cross-model panels (v5 §9), full-text typing, and dual-model computation — all converge on the same verdict against real field outcomes: **no reliable predictive power**. To move past this, a metric needs either (a) a genuinely model-independent signal (the two LLMs' 0.17 agreement says contribution-typing isn't one), or (b) a signal built from the downstream record itself (replication, reuse, stance edges) rather than from a single reading of the paper. Building the metric *from* the field-response graph — not asking an LLM to predict it — is the direction this null points to.

Reproduction

All under `research/metrics/v6-historical/`: `fetch_historical.py` (corpus), `fulltext/` + `ingest_pdfs.py` (full text), `type_ft_claude_workflow.js` + `type_ft_gpt.py` (dual-model typing → `metric_pubtime_ft_{claude,gpt}.json`), `compute_disruption.py` (→ `disruption.json`), `make_overlap_fig.py` (the figure), `RESULTS_ft.md`. All correlations Spearman, recomputed from raw data; ground truth carries no LLM.